

Taking on from the datasets and methodologies that were discussed in the previous chapter, we will now move on to getting a sample set working for object detection. First of all, ensure that you've got a working Jupyter Notebook with extensions for Python and TensorFlow installed on the system.

While you can certainly run Deep Learning frameworks on any compatible hardware, it would be much better if you were to make use of a framework that is optimised to run on that particular hardware. Intel, for example, has optimised versions of popular frameworks such as TensorFlow, MXNET, NEON™, Caffe, etc.

You can refer to the previous chapter to understand the different optimised frameworks that would be best suited for your application. Benchmarks run on Intel hardware to gauge the effect of optimisations indicate an increase in performance up to 70x for training cases and in inference, the improvement was up to 85x. The benchmark used an optimised version of TensorFlow running on Intel® Xeon® and Intel® Xeon Phi™ processor based platforms.

The problem statement

We will be dealing with a model that can identify hardware tools and other elements using TensorFlow. To train the TensorFlow model, there must be two files present that will form the crux of the entire project. A custom object Model files (.pb) and object names file (.pbtxt).

First, install TensorFlow and the Python extension toolkit. To get the full potential of the object detection capabilities, download the Object Detection API from the Google Developer's site.

The API has a full interface available that can provide useful API scripts for training the model. Open up the repository and clone it either from the main website or from GitHub.

To train the model, you will need some training data to work with. You could be creating the program for detecting any sort of object, but the trained API will differ depending on the training model that uses different images.

- apt-get update
- apt-get install -y protobuf-compiler python-pil python-lxml python-pip python-dev git
- pip install Flask==0.12.2 WTForms==2.1 Flask_WTF==0.14.2 Werkzeug==0.12.2
- cd /opt